

Sound / Audio in Multimedia Computing

Basic Sound Concepts, Types of Sound, Digitizing Sound, Audio Formats, Audio Tools, MIDI, Speech Generation, Speech Analysis, and Speech Transmission

Summary

Topic	Description
Basic Sound Concepts	Frequency, amplitude, wavelength, time period
Types of Sound	Analog, digital, audible, infrasonic, ultrasonic
Digitizing Sound	Sampling, quantization, encoding
Audio Formats	WAV, MP3, AAC, FLAC, OGG
Audio Tools	Recording, editing, mixing, analysis software
MIDI	Stores musical performance instructions rather than audio
Speech Generation	Converts text into speech (TTS)
Speech Analysis	Extracts information from speech for recognition and analysis
Speech Transmission	Captures, digitizes, compresses, transmits, and reconstructs speech



5

1. Introduction to Sound (Audio)

Sound is one of the most important components of multimedia. It enhances communication, entertainment, education, virtual reality, gaming, and human-computer interaction.

Without audio, multimedia applications become less interactive and engaging.

Definition

Sound is a form of mechanical energy produced by vibrating objects that travels through a medium (such as air, water, or solids) in the form of longitudinal waves and is perceived by the human ear.

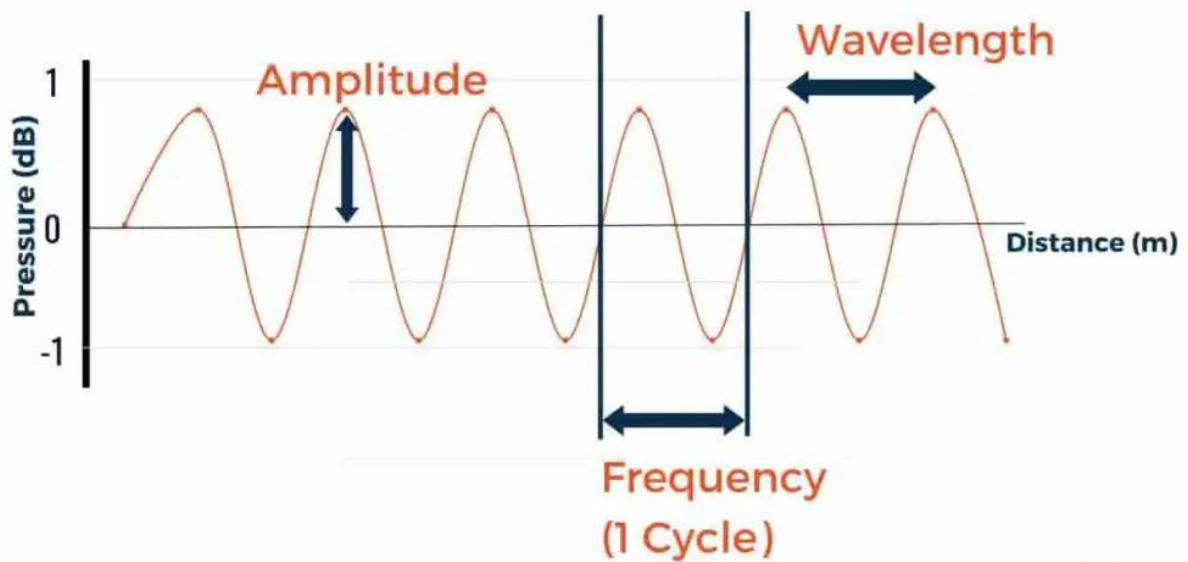
Examples

- Human speech
- Music
- Animal sounds
- Vehicle horns
- Rainfall
- Musical instruments

Characteristics of Sound

Every sound has four important characteristics.

Amplitude & Frequency & Wavelength



Wavelength Comparison

Wavelength	Frequency	Examples
Long	Low	Bass sounds, earthquakes
Medium	Medium	Human speech
Short	High	Ultrasound, bats

Relationship Between Frequency, Amplitude, and Wavelength

Property	Determines	Unit	Effect When Increased
Frequency	Pitch	Hertz (Hz)	Higher pitch
Amplitude	Loudness	Decibel (dB)	Louder sound
Wavelength	Distance between wave peaks	Meter (m)	Longer wavelength corresponds to lower frequency (for a constant wave speed)

Comparison of Frequency, Amplitude, and Wavelength

Feature	Frequency	Amplitude	Wavelength
Definition	Number of vibrations per second	Maximum displacement of the wave	Distance between two consecutive crests or compressions
Symbol	f	A	λ
Unit	Hertz (Hz)	Decibel (dB) for sound level (wave amplitude itself is a displacement)	Meter (m)
Determines	Pitch	Loudness	Spatial length of one wave cycle

Feature	Frequency	Amplitude	Wavelength
High Value Means	High pitch	Loud sound	Long wavelength if wave speed changes, but at constant speed high frequency actually means short wavelength

1. Frequency (Pitch)

Frequency indicates the number of vibrations occurring in one second.

Unit

Hertz (Hz)

Formula

$$f=1/T$$

Where:

- **f** = Frequency (Hz)
- **T** = Time period (seconds)

Example

A sound vibrating **440 times per second** has a frequency of:

440 Hz

This corresponds to the musical note **A4**, commonly used for tuning instruments.

Human Hearing Range

20 Hz – 20,000 Hz (20 kHz)

2. Amplitude (Loudness)

Amplitude represents the maximum displacement of the sound wave.

Higher amplitude produces louder sound.

Unit

Decibel (dB)

Examples

Sound	Approximate Level
Whisper	30 dB
Normal conversation	60 dB
Busy traffic	85 dB
Rock concert	110 dB
Jet engine (close range)	130–140 dB

3. Time Period

Time required to complete one vibration.

$$T=1/f$$

4. Wavelength

Distance between two consecutive wave peaks.

Formula:

$$\lambda=v/f$$

Where

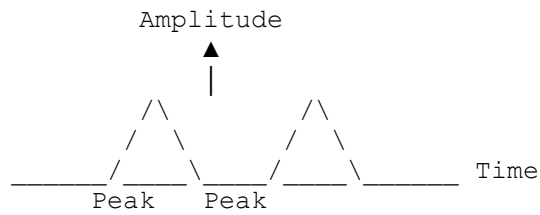
- λ = Wavelength
- v = Speed of sound
- f = Frequency

Speed of sound in air (approximately):

343 m/s

at **20°C**.

Basic Sound Wave



Distance between two peaks = Wavelength (λ)

2. Types of Sound

A. Analog Sound

Analog sound is continuous.

Examples

- Human voice
- Guitar sound
- Piano
- Radio broadcasting

Advantages

- Natural representation

Disadvantages

- Easily affected by noise.
-

B. Digital Sound

Digital sound is represented as binary numbers.

Examples

- MP3
- WAV
- Digital recording
- CDs

Advantages

- Easy storage
 - Noise resistant
 - Easy editing
-

C. Audible Sound

Frequency:

20 Hz–20 kHz

Humans can hear this range.

Examples

- Speech
 - Music
 - Animal sounds
-

D. Infrasonic Sound

Frequency:

Below 20 Hz

Examples

- Earthquakes
 - Volcanoes
 - Elephants (low-frequency communication)
-

E. Ultrasonic Sound

Frequency:

Above 20 kHz

Examples

- Medical ultrasound imaging

- SONAR
- Bats
- Dolphins

Summary of Sound Types

Type	Frequency	Examples
Infrasonic	<20 Hz	Earthquakes, elephant communication
Audible	20 Hz–20 kHz	Speech, music
Ultrasonic	>20 kHz	Ultrasound, bats, SONAR

3. Digitizing Sound

Digitization converts analog sound into digital data.

Digitizing Process

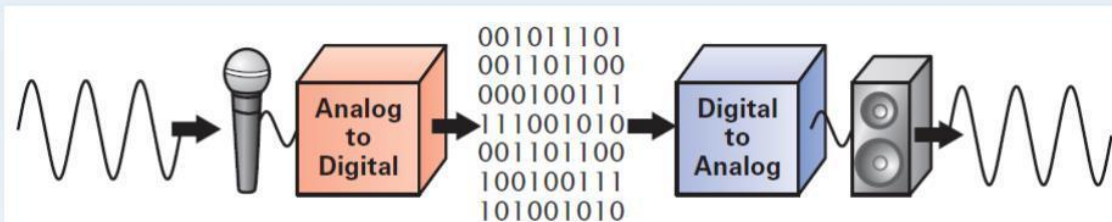


Figure 8.14 Schematic for analog-to-digital and digital-to-analog conversion.

Introduction

Digitizing Sound is the process of converting continuous **analog sound waves** into **digital data (binary numbers)** so that computers can store, process, edit, and transmit audio.

This process is also known as **Analog-to-Digital Conversion (ADC)** and is a fundamental concept in **Multimedia Computing**, **Digital Signal Processing (DSP)**, **Computer Networks**, and **Audio Engineering**.

Digitizing Sound Process (Analog-to-Digital Conversion)

Step 1: Analog audio signal - Continuous representation of signal



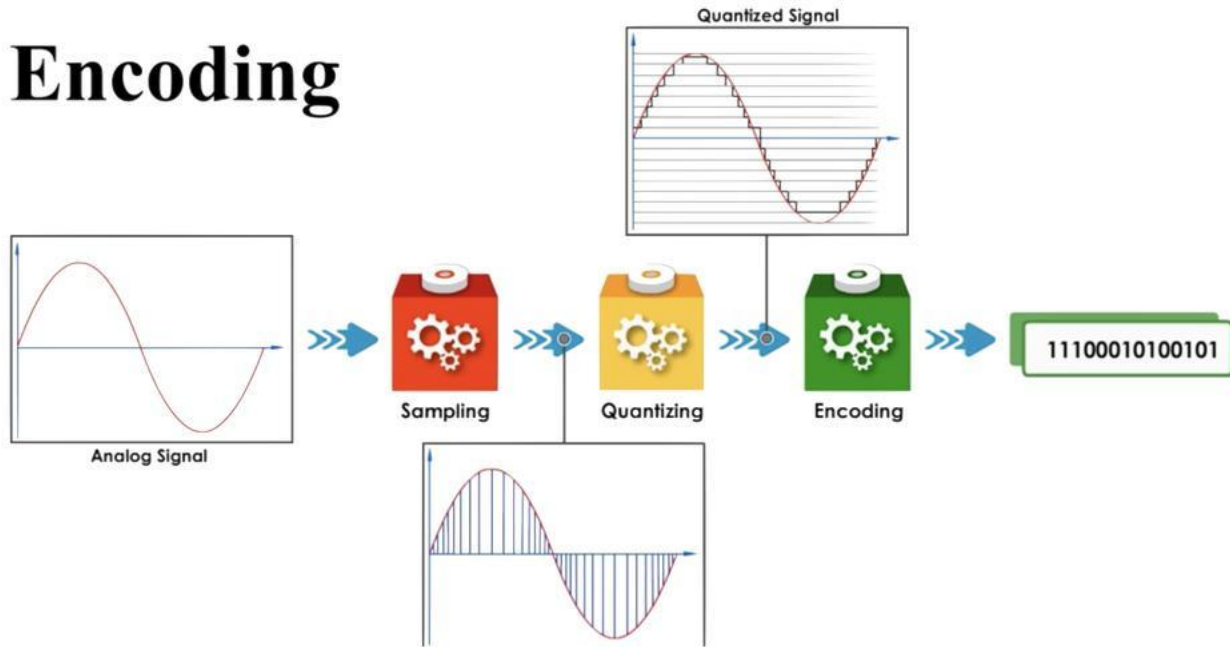
Step 2: Sampling - Samples are selected at regular time intervals



Step 3: Digital audio signal - The way it is stored in memory

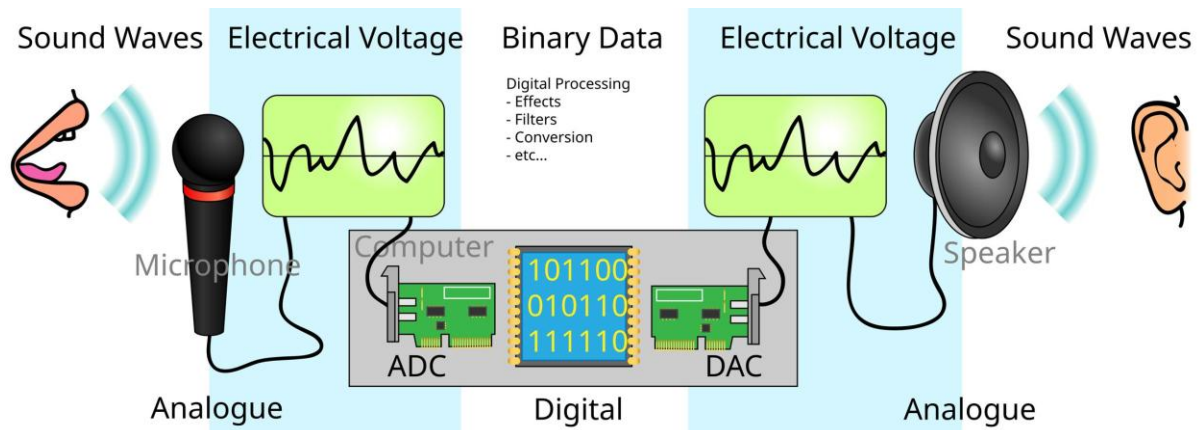


Encoding



Definition

Digitizing Sound is the process of converting analog audio signals into digital form by performing **sound acquisition, sampling, quantization, encoding, compression (optional), storage, and playback.**



Why Do We Need Digitizing Sound?

Analog sound cannot be directly processed by a computer because computers understand only **binary digits (0 and 1).**

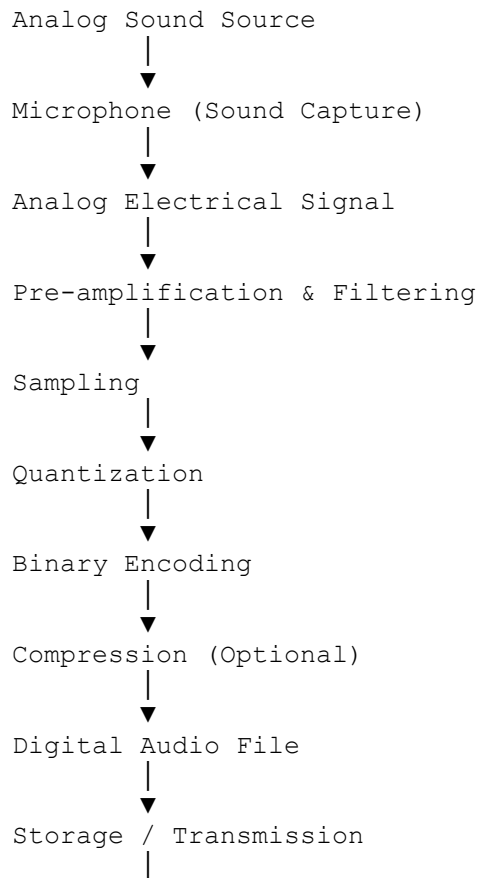
Digitization enables computers to:

- Store audio files.
- Edit and enhance sound.
- Transmit audio over the Internet.
- Stream music and podcasts.
- Recognize speech using Artificial Intelligence.
- Compress audio efficiently.

Real-Life Examples

- Recording voice using a smartphone.
- Making a voice call over the Internet.
- Recording songs in a studio.
- Uploading audio to cloud storage.
- Voice assistants such as Siri, Google Assistant, and Alexa.

Complete Digitizing Sound Process



▼
Playback (Digital-to-Analog Conversion)

The digitization process consists of several important stages.

Summary of Digitizing Sound Process

Stage	Description
1	Sound Generation – Vibrations create sound waves
2	Sound Capture – Microphone converts sound into an analog electrical signal
3	Signal Conditioning – Amplify and filter the signal
4	Sampling – Measure amplitude at regular intervals
5	Quantization – Convert samples into discrete levels
6	Encoding – Represent samples in binary form
7	Compression (Optional) – Reduce file size
8	Storage – Save as a digital audio file
9	Transmission – Send through a communication network
10	Playback – Convert digital data back to analog using a DAC and reproduce it through speakers

Stage 1: Sound Generation

Definition

Sound is produced when an object vibrates.

Examples include:

- Human voice
- Guitar strings
- Piano
- Drum
- Car horn

These vibrations create pressure waves that travel through air.

Fact

Sound cannot travel in a vacuum because it requires a material medium (air, water, or solids).

Stage 2: Sound Capture (Microphone)

Definition

A **microphone** captures sound waves and converts them into an **analog electrical signal**.

Working Principle

Voice

↓

Microphone Diaphragm Vibrates

↓

Electrical Signal Produced

The analog electrical signal has the same shape as the original sound wave.

Types of Microphones

- Dynamic Microphone
- Condenser Microphone
- Ribbon Microphone
- Lavalier (Lapel) Microphone
- USB Microphone

Stage 3: Signal Conditioning (Pre-amplification and Filtering)

Definition

Before digitization, the analog signal is prepared for accurate conversion.

Main Operations

A. Amplification

Weak microphone signals are amplified.

B. Noise Filtering

Unwanted electrical noise is reduced.

C. Anti-Aliasing Filter

A low-pass filter removes frequencies above half the sampling rate (the **Nyquist frequency**) to reduce **aliasing**, where high-frequency components could otherwise appear as incorrect lower frequencies after sampling.

Stage 4: Sampling

Definition

Sampling measures the amplitude of the analog signal at **regular time intervals**.

Instead of recording a continuous waveform, the system stores many discrete samples every second.

Sampling Frequency

Number of samples collected every second.

Unit:

Hertz (Hz)

Mathematical Formula

Sampling Rate=Samples/Second
Sampling Rate=Samples/Second

Example

Sampling Rate:

44,100 Hz

This means

44,100 44,100 44,100

samples are taken every second.

Common Sampling Rates

Sampling Rate	Applications
8 kHz	Telephone systems
16 kHz	Voice recognition
22.05 kHz	Voice recording
44.1 kHz	Audio CD
48 kHz	Video production
96 kHz	Professional recording
192 kHz	Studio mastering

Nyquist Sampling Theorem

To accurately reproduce a signal,

$\text{Sampling Rate} \geq 2 \times \text{Highest Frequency}$
 $\text{Sampling Rate} \geq 2 \times \text{Highest Frequency}$

Example

Human hearing range:

20 kHz

Minimum sampling rate:

$2 \times 20 = 40 \text{ kHz}$ $2 \times 20 = 40 \text{ kHz}$ $2 \times 20 = 40 \text{ kHz}$

Therefore,

44.1 kHz is sufficient for CD-quality audio.

Illustration

Continuous Wave

~~~~~

↓

Sampling

• • • • •

Each dot represents one sample.

---

## Stage 5: Quantization

### Definition

After sampling, each sample is assigned the **nearest available amplitude level**.

## Introduction

**Quantization** is the **second major step** in the digitization of sound, following **sampling**. During quantization, each sampled amplitude of an analog signal is assigned to the nearest available digital value (quantization level). Since computers can only represent a finite number of values, continuous amplitudes must be approximated.

Without quantization, computers would not be able to store analog sound as digital data.

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## Definition

**Quantization** is the process of converting the continuous amplitude values obtained during sampling into a finite set of discrete levels that can be represented digitally.

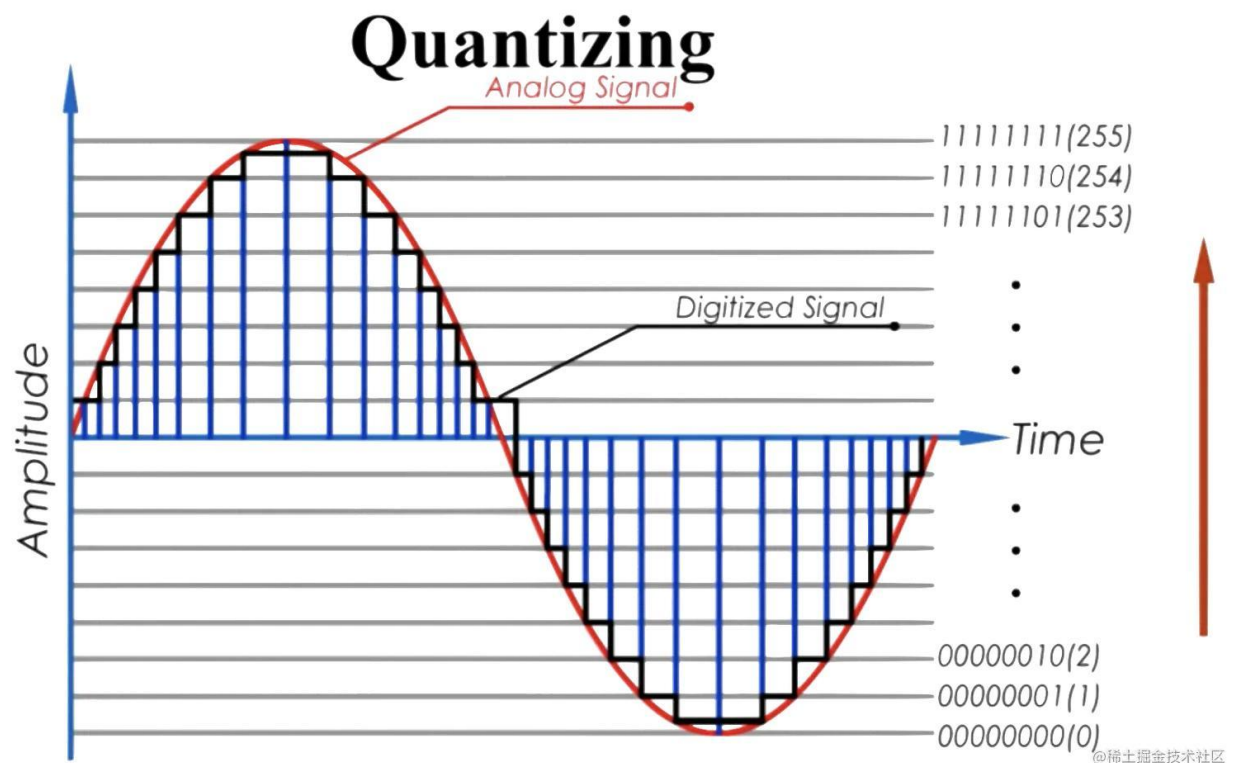
## Applications of Quantization

- Digital music recording
- Audio CDs
- Mobile phones
- Voice over IP (VoIP)
- Speech recognition
- Hearing aids
- Medical audio systems
- Multimedia streaming

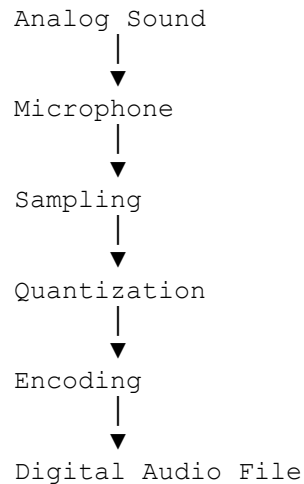
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## Advantages

- Converts analog amplitudes into digital values.
- Enables digital storage and processing.
- Supports efficient transmission.
- Improves compatibility with computer systems.
- Higher bit depths provide more accurate sound representation.



# Position of Quantization in Audio Digitization



## Why is Quantization Necessary?

During **sampling**, the analog signal is measured at regular intervals.

Example sampled values:

0.12  
0.57  
1.43  
2.98  
3.67

These values are continuous and may contain many decimal places.

However, computers can only store a limited number of binary values. Therefore, each sampled value is approximated to the nearest available quantization level.

### Example

Sampled value:

3.67

Nearest available level:

The computer stores **4** instead of **3.67**.

---

# Continuous vs. Discrete Values

## Before Quantization (Continuous)

0.12  
0.35  
1.58  
2.76  
3.91  
4.42

Infinite possible values exist between any two amplitudes.

---

## After Quantization (Discrete)

0  
0  
2  
3  
4  
4

Only fixed amplitude levels are allowed.

Since computers cannot store infinitely many amplitude values, they approximate each sample.

---

## Bit Depth

Bit depth determines how many amplitude levels are available.

Formula

$Levels = 2^n$

Where

n = Bit Depth

---

## Examples

| Bit Depth | Levels                                 |
|-----------|----------------------------------------|
| 8-bit     | 256                                    |
| 16-bit    | 65,536                                 |
| 24-bit    | 16,777,216                             |
| 32-bit    | 4,294,967,296 (integer representation) |

Higher bit depth generally provides better dynamic range and less quantization error.

---

## Quantization Illustration

Original Value = 7.84

↓

Nearest Level

8

This approximation introduces **quantization error**, which is usually reduced by using higher bit depths.

---

## Stage 6: Encoding

### Definition

Each quantized sample is converted into **binary numbers**.

Example

Amplitude:

145

Binary:

10010001

The computer stores these binary values.

---

## Stage 7: Compression (Optional)

### Definition

Compression reduces the size of the digital audio file.

---

### Lossless Compression

Original audio is perfectly reconstructed.

Examples:

- FLAC
- ALAC

Applications

- Music archives
  - Professional editing
- 

### Lossy Compression

Some audio information is discarded to reduce file size.

Examples:

- MP3
- AAC
- OGG Vorbis

Applications

- Music streaming
- Podcasts

- Mobile devices
- 

## Stage 8: Storage

Digital audio is stored in files.

Examples

- WAV
- MP3
- AAC
- FLAC
- OGG

Storage media

- Hard Disk
  - SSD
  - USB Flash Drive
  - CD/DVD
  - Cloud Storage
- 

## Stage 9: Transmission

The audio file may be transmitted through:

- Internet
- Wi-Fi
- Bluetooth
- Mobile Network
- Optical Fiber

Examples

- Voice calls
  - Music streaming
  - Online meetings
  - Podcasts
-

# Stage 10: Playback (Digital-to-Analog Conversion)

## Definition

During playback, the digital samples are converted back into an analog electrical signal by a **Digital-to-Analog Converter (DAC)**.

## Process

Digital File

↓

DAC

↓

Amplifier

↓

Speaker

↓

Sound Heard

The speaker converts the electrical signal into sound waves that the listener hears.

---

## Mathematical Example

Suppose

Sampling Rate:

44,100 Hz

Bit Depth:

16 bits

Stereo Audio

Number of Channels:

2

Duration:

60 seconds

### Calculate Storage Requirement

$$\begin{aligned}\text{Size} &= \text{Sampling Rate} \times \text{Bit Depth} \times \text{Channels} \times \text{Time Size} \\ &= 44100 \times 16 \times 2 \times 60 = 44100 \times 16 \times 2 \times 60 \\ &= 84,672,000 \text{ bits} = 84,672,000 \text{ bits}\end{aligned}$$

Convert into bytes

$$\begin{aligned}84,672,000 \text{ bits} &= \frac{84,672,000}{8} = 10,584,000 \text{ bytes} \\ &= 10,584,000 \text{ bytes}\end{aligned}$$

$\approx 10.1 \text{ MB}$  (using 1 MB = 1,048,576 bytes, the value is approximately 10.1 MiB).

---

## Real-Life Example: Recording a Song

Suppose a singer records a song.

### Step 1

Singer sings.

↓

### Step 2

Microphone captures sound.

↓

### Step 3

Audio interface amplifies and filters the signal.

↓

#### **Step 4**

ADC samples the waveform (e.g., 44.1 kHz).



#### **Step 5**

Samples are quantized (e.g., 24-bit).



#### **Step 6**

Samples are encoded into binary.



#### **Step 7**

The recording is saved as WAV.



#### **Step 8**

The final version is compressed to MP3 or AAC for online streaming.

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## **Advantages of Digitizing Sound**

- High audio quality.
  - Easy editing and processing.
  - Efficient storage and compression.
  - Reliable transmission.
  - Supports AI-based speech recognition.
  - Minimal quality loss when copying digital files.
- 

## **Limitations**

- Higher sampling rates and bit depths require more storage.

- Lossy compression may reduce audio quality.
  - Background noise affects recordings.
  - Real-time processing requires sufficient computing resources.
- 

## Summary of Digitizing Sound Process

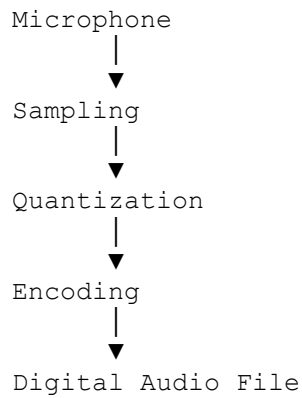
| Stage | Description                                                                                  |
|-------|----------------------------------------------------------------------------------------------|
| 1     | Sound Generation – Vibrations create sound waves                                             |
| 2     | Sound Capture – Microphone converts sound into an analog electrical signal                   |
| 3     | Signal Conditioning – Amplify and filter the signal                                          |
| 4     | Sampling – Measure amplitude at regular intervals                                            |
| 5     | Quantization – Convert samples into discrete levels                                          |
| 6     | Encoding – Represent samples in binary form                                                  |
| 7     | Compression (Optional) – Reduce file size                                                    |
| 8     | Storage – Save as a digital audio file                                                       |
| 9     | Transmission – Send through a communication network                                          |
| 10    | Playback – Convert digital data back to analog using a DAC and reproduce it through speakers |

## Conclusion

The **Digitizing Sound Process** converts continuous analog sound into digital audio that computers can store, process, transmit, and reproduce. It involves **sound capture, signal conditioning, sampling, quantization, encoding, optional compression, storage, transmission, and playback**. High-quality digital audio depends primarily on **appropriate sampling rates, sufficient bit depth, and suitable compression methods**, making this process essential for modern multimedia systems, digital music, speech recognition, telecommunications, and audio streaming.

## Digitizing Process

Analog Sound  
↓  
▼



## Step 1: Sampling

The analog signal is measured at regular intervals.

### Sampling Rate

Number of samples collected per second.

### Unit

Samples/second (Hz)

### Common Sampling Rates

| Sampling Rate | Applications           |
|---------------|------------------------|
| 8 kHz         | Telephone              |
| 22.05 kHz     | Voice recording        |
| 44.1 kHz      | Audio CD               |
| 48 kHz        | Video production       |
| 96 kHz        | Professional recording |
| 192 kHz       | Studio mastering       |

### Nyquist Theorem

To reproduce a signal accurately,

$\text{Sampling Rate} \geq 2 \times \text{Highest Frequency}$

Example

Human hearing = 20 kHz

Minimum sampling rate:

$2 \times 20 = 40 \text{ kHz}$

This is why **44.1 kHz** is sufficient for CD-quality audio.

---

## Step 2: Quantization

Each sampled value is assigned to the nearest discrete amplitude level.

Bit depth determines the number of levels.

Examples

|                                         | Bit Depth                              | Levels     |
|-----------------------------------------|----------------------------------------|------------|
| 8-bit                                   |                                        | 256        |
| 16-bit                                  |                                        | 65,536     |
| 24-bit                                  |                                        | 16,777,216 |
| 32-bit (floating point in many systems) | Very high dynamic range for processing |            |

Higher bit depth generally provides greater dynamic range and lower quantization noise.

---

## Step 3: Encoding

Quantized values are stored as binary numbers.

Example

```
10100111
01101001
11001010
```

---

# 4. Audio Formats

Audio formats store digital sound.

## **A. WAV (Waveform Audio File Format)**

### Characteristics

- Uncompressed
- High quality
- Large file size

### Applications

- Studio recording
  - Audio editing
- 

## **B. MP3 (MPEG Audio Layer III)**

### Characteristics

- Lossy compression
- Small file size

### Applications

- Music streaming
  - Portable music players
- 

## **C. AAC (Advanced Audio Coding)**

### Characteristics

- Better compression efficiency than MP3 at similar bitrates

### Applications

- Streaming services
  - Mobile devices
- 

## **D. FLAC (Free Lossless Audio Codec)**

### Characteristics

- Lossless compression
- High quality
- Smaller than WAV

### Applications

- Music archiving
  - Audiophile collections
- 

## E. OGG Vorbis

### Characteristics

- Open-source
- Good quality
- Efficient compression

### Applications

- Games
  - Streaming
- 

## Audio Format Comparison

|      | Format   | Compression | Quality   | File Size  |
|------|----------|-------------|-----------|------------|
| WAV  | None     |             | Excellent | Very Large |
| MP3  | Lossy    |             | Good      | Small      |
| AAC  | Lossy    |             | Very Good | Small      |
| FLAC | Lossless |             | Excellent | Medium     |
| OGG  | Lossy    |             | Good      | Small      |

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## 5. Audio Tools

Audio tools are software applications used to create, edit, record, mix, and analyze audio.

### Examples

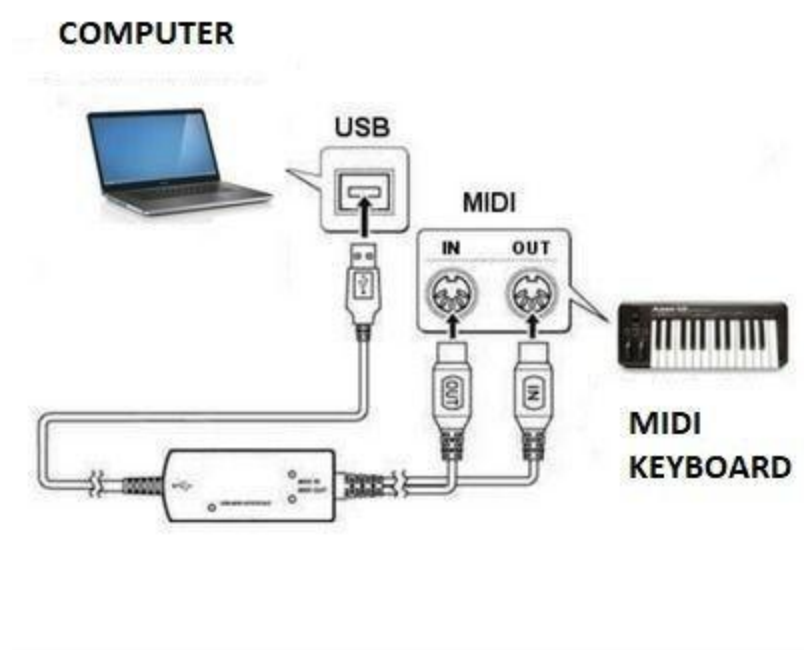
- Audacity
- Adobe Audition
- GarageBand
- Reaper
- Ocenaudio
- Logic Pro
- Pro Tools

## Functions

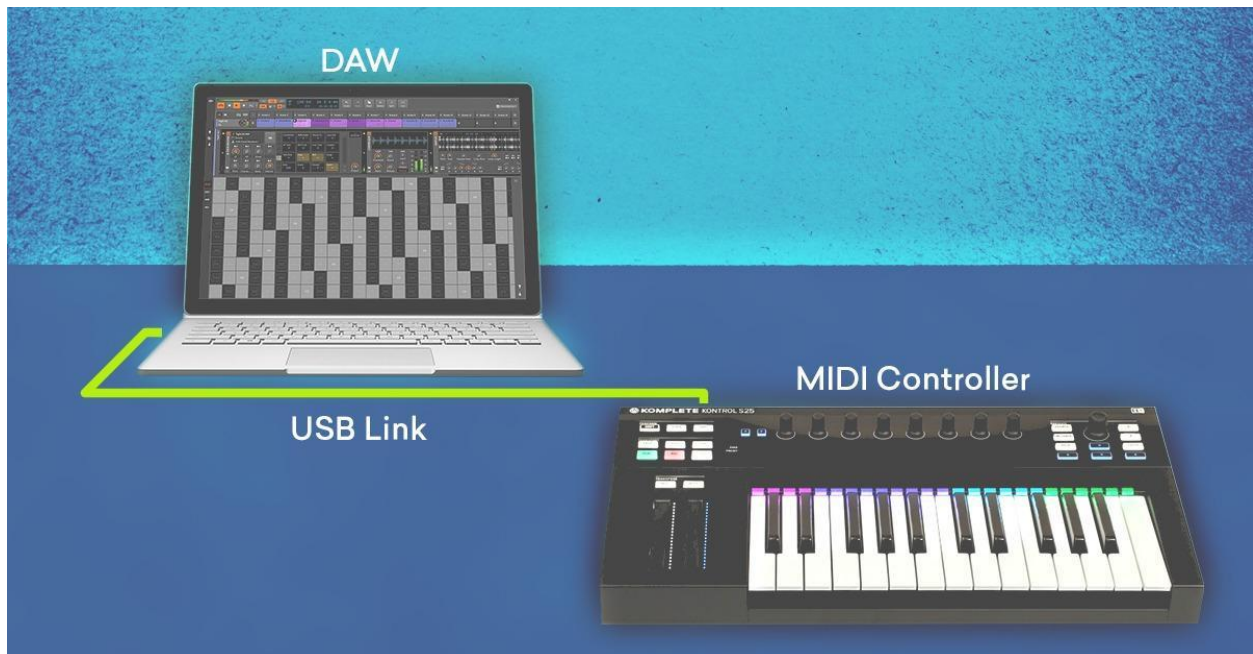
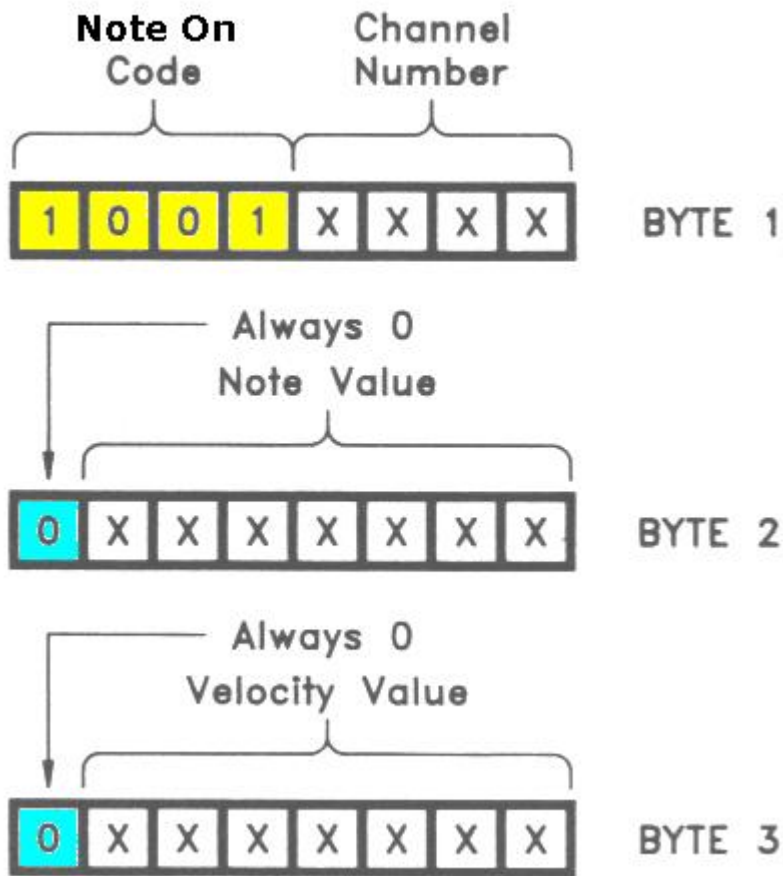
- Recording
- Editing
- Noise reduction
- Mixing
- Equalization
- Audio effects
- Format conversion
- Podcast production

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## 6. MIDI (Musical Instrument Digital Interface)



## NOTE ON MESSAGE



# Definition

**MIDI (Musical Instrument Digital Interface)** is a communication standard that allows electronic musical instruments, computers, and other devices to exchange musical performance information.

Unlike MP3 or WAV, MIDI stores **performance instructions**, not recorded sound.

## Example

Instead of storing:

"A piano sound"

MIDI stores:

- Piano instrument
- Note C
- Volume
- Duration
- Tempo

The receiving device generates the sound.

---

## Components

- MIDI Keyboard
  - MIDI Controller
  - Computer
  - MIDI Interface
  - Synthesizer
- 

## MIDI Messages

Examples

- Note ON
- Note OFF
- Pitch Bend
- Program Change
- Control Change

- Velocity
- 

## Advantages

- Very small file size
  - Easy editing
  - Instrument changes without rerecording
  - Ideal for music composition
- 

## Limitations

- Depends on the synthesizer or sound library for playback quality.
  - Not suitable for recording human speech.
- 

# 7. Speech Generation

## Definition

Speech generation is the process of converting text into spoken language.

Also known as

**Text-to-Speech (TTS)**

## Introduction

**Speech Generation** is the process of converting written text into natural, understandable speech using computer technology. It enables computers and smart devices to "speak" by transforming text into audio signals.

Speech generation is also called **Text-to-Speech (TTS)** or **Speech Synthesis**.

It is a fundamental technology in **Multimedia Computing**, **Artificial Intelligence (AI)**, **Natural Language Processing (NLP)**, **Human-Computer Interaction (HCI)**, and **Assistive Technology**.

---

# Definition

**Speech Generation (Text-to-Speech)** is the process of automatically converting digital text into intelligible and natural-sounding speech using linguistic analysis, pronunciation modeling, prosody generation, and speech synthesis techniques.

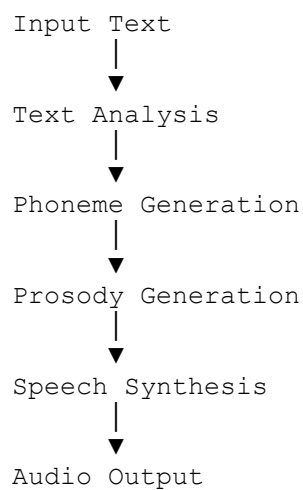
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## Objectives of Speech Generation

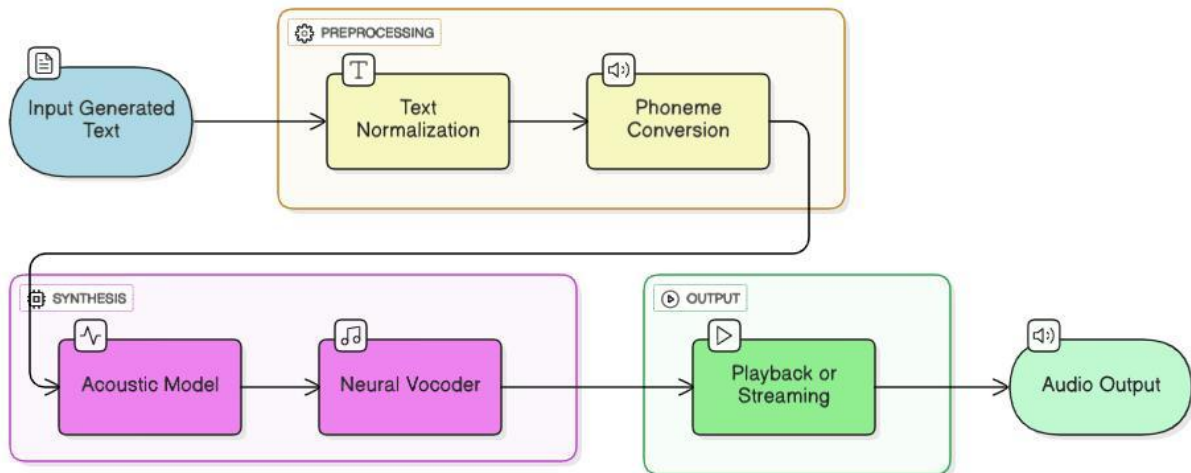
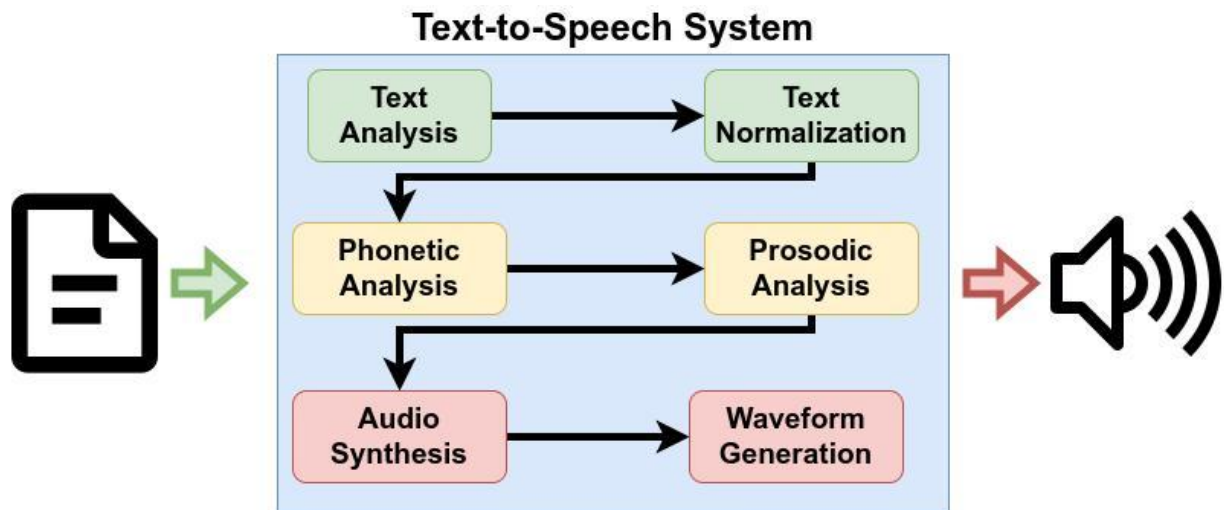
The primary goals are to:

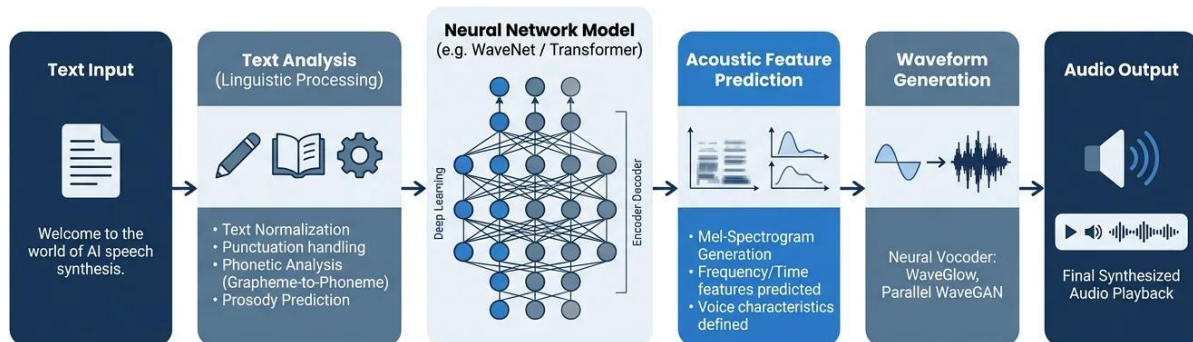
- Convert text into spoken language.
  - Produce natural and intelligible speech.
  - Improve accessibility for visually impaired users.
  - Enable voice assistants and conversational systems.
  - Support multilingual communication.
  - Provide audio content for education and entertainment.
- 

## Speech Generation Process

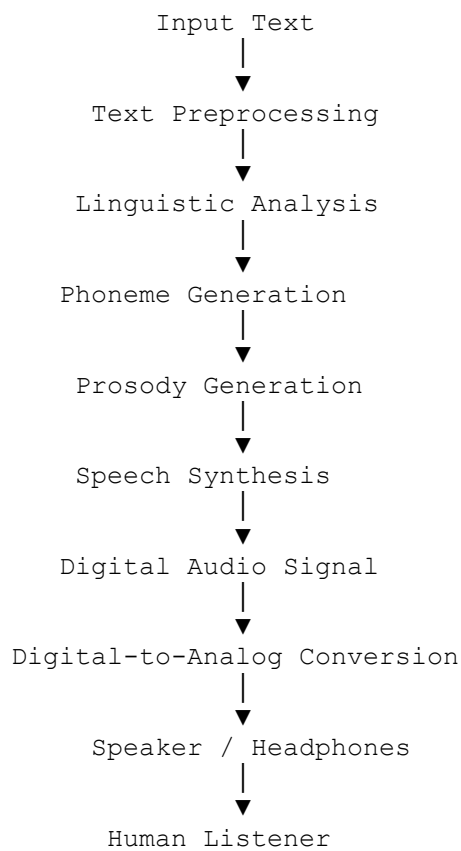


## Speech Generation Process (Text-to-Speech – TTS)





## Complete Speech Generation Process



The speech generation process consists of several important stages.

# Summary of Speech Generation Process

| Stage | Description                                                                               |
|-------|-------------------------------------------------------------------------------------------|
| 1     | Text Input – Receive digital text from the user or application                            |
| 2     | Text Preprocessing – Normalize numbers, abbreviations, dates, symbols, and punctuation    |
| 3     | Linguistic Analysis – Perform tokenization, grammatical, syntactic, and semantic analysis |
| 4     | Phoneme Generation – Convert words into pronunciation units (phonemes)                    |
| 5     | Prosody Generation – Add pitch, stress, rhythm, pauses, and intonation                    |
| 6     | Speech Synthesis – Generate a speech waveform using synthesis techniques                  |
| 7     | Digital Audio Generation – Produce digital audio samples                                  |
| 8     | Digital-to-Analog Conversion – Convert digital samples into an analog signal              |
| 9     | Speaker Output – Play the speech through speakers or headphones                           |

## Stage 1: Text Input

### Definition

The process begins with digital text entered by a user or obtained from a document, webpage, or application.

### Examples

Welcome to Multimedia Computing.

or

Today's weather is sunny.

The system first receives this text as input.

---

## Stage 2: Text Preprocessing (Normalization)

### Definition

Raw text often contains abbreviations, numbers, dates, symbols, and punctuation that cannot be spoken directly. Text preprocessing converts them into a form suitable for speech.

## Main Activities

### A. Remove unnecessary symbols

Example

Hello!!!

↓

Hello

---

### B. Expand abbreviations

#### Abbreviation Spoken Form

|       |           |
|-------|-----------|
| Dr.   | Doctor    |
| Mr.   | Mister    |
| Prof. | Professor |
| etc.  | et cetera |

Example

Dr. Sharma

↓

Doctor Sharma

---

### C. Convert numbers

Example

2026

↓

Two thousand twenty-six

---

### D. Convert dates

Example

08/07/2026



Eighth of July, two thousand twenty-six

---

## **E. Convert currency**

Example

Rs. 500



Five hundred rupees

---

## **F. Handle punctuation**

Punctuation influences pauses and intonation.

Example

Hello, everyone.

The comma creates a short pause, while the period marks the end of the sentence.

---

# **Stage 3: Linguistic Analysis**

## **Definition**

The system analyzes the grammatical and linguistic structure of the sentence.

## **Main Activities**

### **A. Tokenization**

The sentence is divided into words and punctuation.

Example

Welcome to Multimedia Computing.



Welcome  
to  
Multimedia  
Computing

---

## B. Part-of-Speech (POS) Tagging

Each word is identified according to its grammatical role.

Example

| Word       | POS                         |
|------------|-----------------------------|
| Welcome    | Verb                        |
| Multimedia | Adjective (in this context) |
| Computing  | Noun                        |

---

## C. Syntactic Analysis

The sentence structure is analyzed to understand relationships between words.

---

## D. Semantic Analysis

The system interprets the meaning of the sentence to improve pronunciation and intonation.

---

# Stage 4: Phoneme Generation

## Definition

A **phoneme** is the smallest unit of sound that distinguishes one word from another in a language.

The system converts words into phoneme sequences using pronunciation dictionaries and language rules.

## Example

Word

Computer



Approximate phonemes

/k ə m 'p ju: t ə r/

Another example

Hello



/h ə 'l oʊ/

---

## Grapheme-to-Phoneme (G2P) Conversion

Unknown words or names are converted into phonemes using pronunciation rules.

Example

ChatGPT



Pronunciation is generated according to language-specific rules.

---

## Stage 5: Prosody Generation

### Definition

**Prosody** refers to the rhythm, stress, pitch, speaking rate, and pauses in speech.

Without prosody, speech sounds robotic.

### Components

#### A. Pitch

Determines whether the voice sounds high or low.

---

## **B. Stress**

Certain syllables are emphasized.

Example

PREsent (noun)

preSENT (verb)

---

## **C. Intonation**

Pitch changes across a sentence.

Example

Statement

You are coming.

Question

Are you coming?

The question usually ends with a rising intonation.

---

## **D. Speaking Rate**

Controls how fast or slowly the speech is produced.

---

## **E. Pause Generation**

The system inserts pauses after commas, periods, and other punctuation marks.

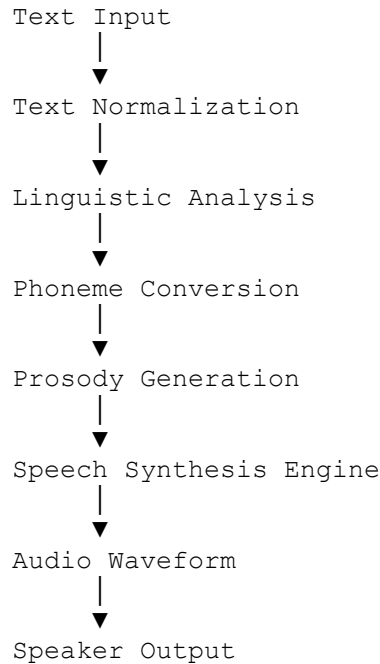
---

# **Stage 6: Speech Synthesis**

## Definition

Speech synthesis converts phonemes and prosody into a digital speech waveform.

## Overall Speech Synthesis Process



## Applications of Speech Synthesis

- Virtual assistants (e.g., Siri, Google Assistant, Alexa)
- GPS navigation systems
- Screen readers for visually impaired users
- Audiobook generation
- Customer service chatbots
- Language learning applications
- Smart home devices
- Automated announcements in transportation systems

## Stage 7: Digital Audio Generation

The synthesized speech is stored as digital audio samples.

Typical properties

| Property      | Example                     |
|---------------|-----------------------------|
| Sampling Rate | 16 kHz, 22.05 kHz, 44.1 kHz |
| Bit Depth     | 16-bit                      |
| Channels      | Mono or Stereo              |

The result is a digital audio stream.

---

## Stage 8: Digital-to-Analog Conversion (DAC)

The digital audio is converted into an analog electrical signal.

Process

Digital Audio

↓

DAC

↓

Analog Signal

---

## Stage 9: Speaker Output

The analog electrical signal is amplified and sent to:

- Speakers
- Headphones
- Mobile phone speakers
- Smart speakers

The listener hears the synthesized speech.

---

## Real-Life Example

Suppose a user enters:

Good morning, everyone.

## Step 1

Input Text



Good morning, everyone.

---

## Step 2

Text Normalization

The text is cleaned and punctuation is interpreted.

---

## Step 3

Linguistic Analysis

Words and sentence structure are analyzed.

---

## Step 4

Phoneme Generation

The words are converted into phonemes.

---

## Step 5

Prosody

The system determines:

- Pitch
  - Stress
  - Pause after the comma
  - Natural intonation
-

## **Step 6**

Speech Synthesis

A speech waveform is generated.

---

## **Step 7**

Digital Audio

The waveform is stored digitally.

---

## **Step 8**

DAC

Digital audio is converted to an analog signal.

---

## **Step 9**

Speaker

The user hears:

"Good morning, everyone."

---

# **Applications of Speech Generation**

## **Education**

- E-learning
- Audiobooks
- Language learning

## **Healthcare**

- Communication aids
- Voice assistance for patients

## **Banking**

- Voice banking
- Automated customer support

## **Navigation**

- GPS voice guidance

## **Smart Devices**

- Voice assistants
- Smart home systems

## **Accessibility**

- Screen readers
- Reading assistance for visually impaired users

## **Entertainment**

- Games
  - Movies
  - Virtual characters
- 

# **Advantages**

- Makes digital content accessible.
  - Provides hands-free interaction.
  - Supports multilingual communication.
  - Produces consistent speech output.
  - Reduces the need for human voice recordings in many applications.
- 

# **Limitations**

- Pronouncing uncommon names or newly coined words can be challenging.
- Emotional expression may still be limited compared with human speech.

- Performance depends on language resources and training data.
  - High-quality neural speech synthesis requires significant computational resources.
- 

# Comparison of Speech Synthesis Techniques

| Technique              | Method                                        | Advantages                                           | Limitations                                          |
|------------------------|-----------------------------------------------|------------------------------------------------------|------------------------------------------------------|
| Concatenative          | Joins prerecorded speech segments             | Very natural for available recordings                | Large storage requirements, limited flexibility      |
| Formant                | Mathematical model of the vocal tract         | Small memory usage, flexible                         | Less natural sounding                                |
| Statistical Parametric | Statistical models generate speech parameters | Consistent and adaptable                             | Can sound less expressive than modern neural systems |
| Neural                 | Deep learning models generate speech          | Highly natural, expressive, supports multiple voices | Requires more computing power and training data      |

## Conclusion

The **Speech Generation Process** transforms written text into natural-sounding spoken language through a sequence of stages: **text input, preprocessing, linguistic analysis, phoneme generation, prosody generation, speech synthesis, digital audio generation, digital-to-analog conversion, and speaker output**. Modern AI-based speech synthesis systems use deep learning to produce highly intelligible and expressive speech, making them essential for multimedia applications, virtual assistants, accessibility technologies, navigation systems, education, and customer service.

## Applications

- GPS navigation
  - Screen readers
  - Virtual assistants
  - Educational software
  - Customer service systems
-

# 8. Speech Analysis

## Definition

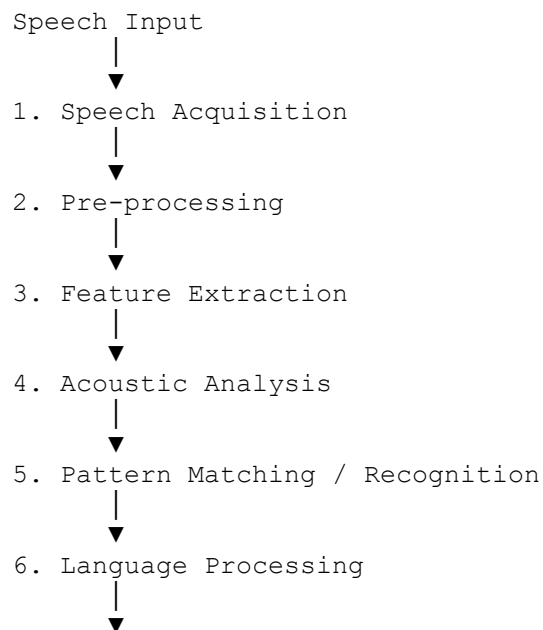
Speech analysis extracts useful information from speech signals.

**Speech Analysis** is the process of extracting meaningful information from a speech signal so that a computer can recognize, interpret, or process spoken language. It is the first stage of an **Automatic Speech Recognition (ASR)** system.

**Speech Analysis** is the process of analyzing a speech signal to extract useful information for recognizing spoken language. The main stages are:

1. **Speech Acquisition** – Capture speech using a microphone and convert it into a digital signal.
2. **Pre-processing** – Remove noise, normalize the signal, and divide it into frames.
3. **Feature Extraction** – Extract important speech features such as MFCC or LPC.
4. **Acoustic Analysis** – Match features with acoustic models to identify phonemes.
5. **Pattern Matching/Recognition** – Recognize words using HMMs, neural networks, or other recognition techniques.
6. **Language Processing** – Apply language rules to produce meaningful and grammatically correct text.

## Stages of Speech Analysis



## Applications

- Voice assistants
  - Speaker identification
  - Emotion detection
  - Language recognition
  - Voice authentication
- 

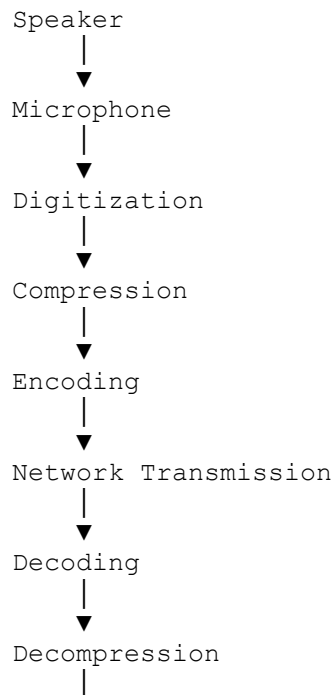
# 9. Speech Transmission

## Definition

Speech transmission is the process of sending speech signals from one location to another over a communication network.

---

## Process



## Applications

- Telephone systems
  - Video conferencing
  - Voice over IP (VoIP)
  - Online education
  - Telemedicine
- 

## Real-Life Example: Online Meeting

When you speak during an online meeting:

1. Your voice is captured by the microphone.
  2. It is digitized through sampling, quantization, and encoding.
  3. The audio is compressed using an audio codec.
  4. The compressed data is transmitted over the Internet.
  5. The receiver decodes and decompresses the data.
  6. The other participant hears your voice through speakers or headphones.
- 

## Advantages of Digital Audio

- High-quality sound.
  - Easy editing and processing.
  - Efficient storage and compression.
  - Fast transmission over networks.
  - Reduced degradation compared to analog copying.
- 

## Limitations

- High-quality recordings require more storage.
- Compression may reduce quality when lossy codecs are used.
- Background noise affects recording quality.
- Real-time communication depends on network bandwidth and latency.

---

# Summary

| Topic                | Description                                                         |
|----------------------|---------------------------------------------------------------------|
| Basic Sound Concepts | Frequency, amplitude, wavelength, time period                       |
| Types of Sound       | Analog, digital, audible, infrasonic, ultrasonic                    |
| Digitizing Sound     | Sampling, quantization, encoding                                    |
| Audio Formats        | WAV, MP3, AAC, FLAC, OGG                                            |
| Audio Tools          | Recording, editing, mixing, analysis software                       |
| MIDI                 | Stores musical performance instructions rather than audio           |
| Speech Generation    | Converts text into speech (TTS)                                     |
| Speech Analysis      | Extracts information from speech for recognition and analysis       |
| Speech Transmission  | Captures, digitizes, compresses, transmits, and reconstructs speech |

## Conclusion

Sound is a core component of multimedia systems that enables effective communication and immersive user experiences. Understanding **basic sound concepts, types of sound, digitization, audio formats, audio tools, MIDI, speech generation, speech analysis, and speech transmission** is essential for designing multimedia applications such as music production, streaming platforms, virtual assistants, online education, telemedicine, gaming, and communication systems. Modern digital audio technologies provide high-quality, efficient, and reliable methods for recording, processing, storing, and transmitting sound across a wide range of applications.